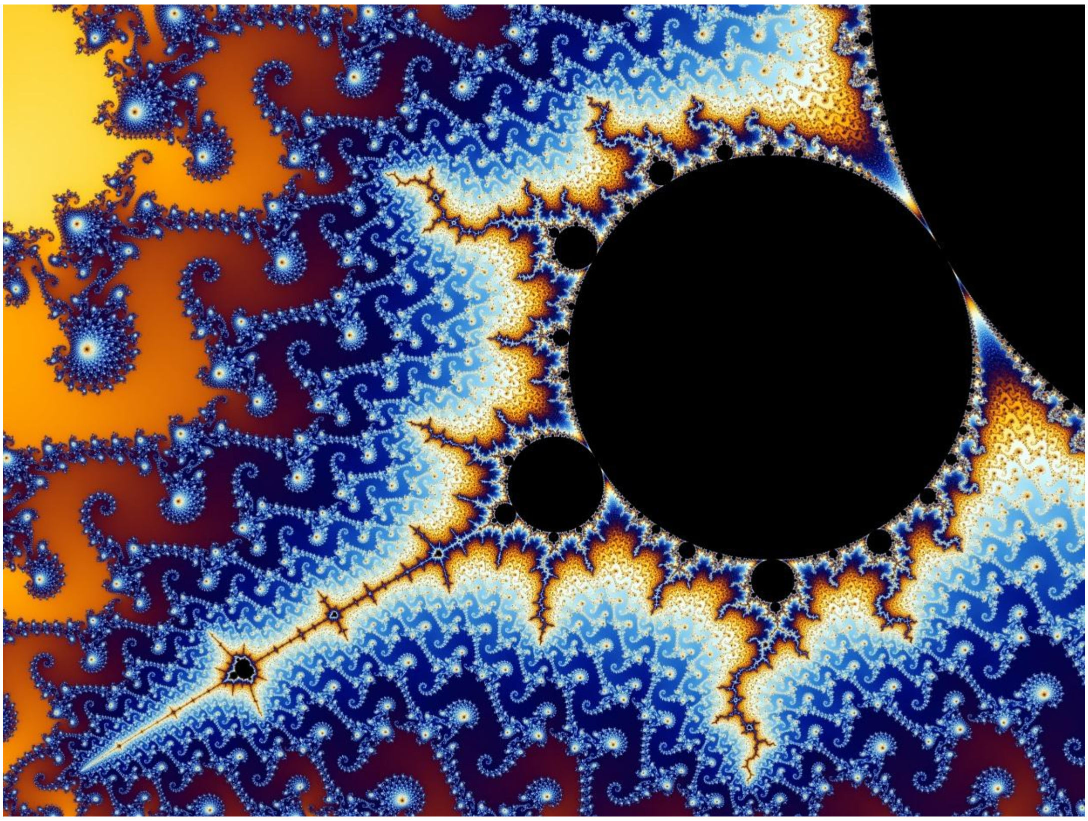
Tour de Maths 2023

Leg 4

Question Paper



The Mandelbrot set

5 Mark Questions

Question 1

How many numbers below one hundred are divisible by both 2 and 3?

Question 2

Sihle has 27 cards. This is three more than four times the number of cards his brother has. If Sihle's brother has x cards, write down an equation to represents this information?

Question 3

The ratio of cars to motorbikes in a car park is 10 cars to 4 motorbikes. There are 70 motorbikes in the car park. How many cars are there?

Question 4

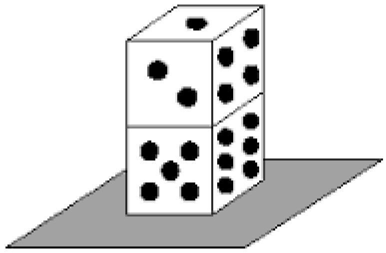
When I was 14 years old my mother was 42 years old, which was three times my age. Now she is twice my age, how old am I?

Question 5

A tortoise and a hare race against each other. A hare runs at a constant speed of 36 km per hour for exactly ten seconds and waits for the tortoise to catch up. If the tortoise takes two hours to move 1 km, how long will it take to catch up?

Question 6

Two ordinary six-sided dice are stacked on top of each other and placed on a tabletop.



What is the sum of the dots on all the visible faces?

Question 7

Mr. and Mrs. Dlamini have two daughters and three sons. At Easter time every member of the family buys one chocolate Easter egg foe each other member. How many Easter eggs will be bought in total?

Question 8

The digits 1, 2, 3, 4, and 9 are each used once to form the smallest possible even five-digit number. What is the digit in the tens place?

Question 9

Three solid balls, each of radius $3{,}25$ cm, are stored in a cylindrical can with the smallest possible radius and volume. What fraction of the can's volume is air?

Question 10

Three angles of a pentagon have measure 88^0, 124^0 and 92^0. If the measures of the remaining two angles are equal, what is the measure, in degrees, of one of the remaining angles?

10 Mark Questions

Question 11

Before he took his last exam, the average of John's exam scores was 89. He figures that if he scores 97 on the last exam, his average for all the exams will be 90. If he scores a 73, his average for all the exams will be 87. How many exams, including the last one, are given in the class?

Question 12

Real numbers *a* and *b* satisfy the equations $3^a = 81^{b+2}$ and $125^b = 5^{a-3}$. What is *ab*?

Question 13

The number 13 is prime and so too is its reverse, 31. How many two-digit primes can you find for which their reverse is also prime?

Question 14

Find the value of the digit c in the following calculation.

$$\begin{array}{r} a\ b \\ -\,b\ a \\ \hline c\ 4 \end{array}$$

Question 15

Let a = 1, b =2 and c = 3, … , z =26. What is the exact value of $((x-a)(x-b)\ldots(x-z)$?

20 Mark Questions

Question 16

The rule of an obstacle course specifies that at the *nth* obstacle a person has to toss a die *n* times. If the sum of points in these tosses is bigger than 2^n, the person is said to have crossed the obstacle. At most how many obstacles can a person cross?

Question 17

Sarah started school at the age of five. She spent one quarter of her life being educated and went straight to work. After working for one half of her life, she lived for fourteen happy years after retiring. How old was she when she retired?

Question 18

A cube measuring 3 x 3 x 3 is made up of 27 smaller cubes and has a 1 x 1 square cube hole pushed through the centre of each face so that you can see straight through the cube from every side.

The number of small cubes remaining is 20. If a 5 x 5 x5 cube has 3 x 3 square holes pushed through the centre of each face, how many smaller cubes would remain?

Question 19

In a group of 35 children, 10 have blonde hair, 14 have brown eyes, and 4 have blonde hair and brown eyes. What fraction of the class hair blonde hair or brown eyes?

Question 20

A train travelling between two towns arrives ten minutes late when travelling at 40 km/h and sixteen minutes late when travelling at 30 km/h. What is the distance between the two towns?

Total: 200

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